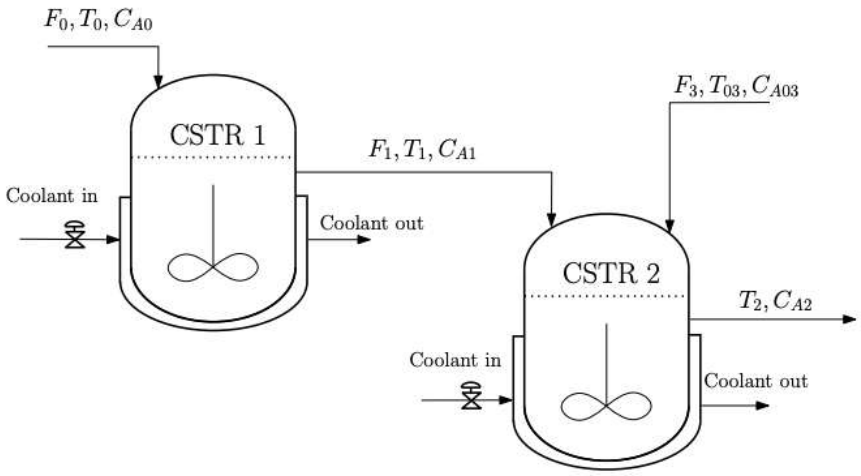
Problem Statement(2-4 sentences)

We are particularly interested in modeling two dynamical continuously stirred tank reactors (CSTR) in series with an exothermic reaction and non-isothermal reactions. This is an interesting system because the ODE system is coupled (mass/mole and energy balances) and depending on process parameters such as reactor sizes, flows, and reaction kinetics, the system can be stiff. To align with our interests, we will assess a stiff system which would require us to analyze the requirements of convergence for numerical ODE approaches.

Figure 1: CSTRs in series configuration (Figure from Knudesen et al.)



Mathematical approach (1/2 page)

We will begin with first-principle modeling. This requires four ODEs as we will have to develop one mass and one temperature balance equation per CSTR. Given that the system is dynamic, changes in temperature affect the mole balance through the Arrhenius equation.

Since the reactors are in series with no recycle, the coupling between them is one-way: Reactor 2 depends on the outlet of Reactor 1, but Reactor 1 is independent of Reactor 2. This means the system can either be solved sequentially (Reactor 1 first, then Reactor 2) or simultaneously as a single system of four ODEs.

The next step is to establish and analyze approaches to solve this system for different priorities. To do this, we must consider three objectives: convergence (order of accuracy and numerical stability), accuracy (error relative to a tight-tolerance reference solution), and computational cost (function evaluations and runtime).

We limit the study to explicit and implicit methods from four families. Euler's method is the simplest case: forward Euler (explicit) and backward Euler (implicit). Runge-Kutta (RK) methods are one-step methods, with explicit RK (e.g., RK4, RK45) and implicit RK (e.g., Radau). Linear multistep (LMS) methods reuse previous steps, with Adams-Bashforth (explicit) and Adams-Moulton (implicit). Finally, backward differentiation formulas (BDF) are implicit LMS methods that are stiffly stable, so we expect them to handle the stiffness from the Arrhenius coupling best. Explicit methods are cheap per step, but their step size is limited by stability on stiff dynamics. Implicit methods cost more per step because they solve a nonlinear system, but they can take much larger steps. Comparing the two groups on this system will show which approach to choose when a faster solve or a more accurate solve is the priority.

Resources needed (1/2 page)

We need to find a suitable reaction with reaction rates that fall into the "moderately" or "very" stiff categories, with the stiffness ratio S > 104. One possible option for this is a simplified version of the polyol production reaction detailed by Yunhan et al. From a found reaction we should be able to generate any data that we need for subsequent calculations, such as predictive modelling of the system under input changes. Software we're planning on using: NumPy, SciPy, and Matplotlib for simulation, continuation, and analysis. Pyomo (with the IPOPT solver) to formulate the steady-state problem as a square nonlinear system and, if time allows, to optimize operating conditions subject to stability constraints. We will install IPOPT via conda, and if installation proves difficult, we will fall back to scipy.optimize_root for steady-state solves. Papers we're basing the project on:

[1] Wen, Y.; Biegler, Lorenz. T.; Ochoa, M. P.; Matthews, L.; Ferrio, J.; Weston, J.; Nikbin, N. Continuous Reactor Network Design for Rigid Polyol Production. Chemical Engineering Science 2021, 230, 116189. https://doi.org/10.1016/j.ces.2020.116189

[2] Knudsen, B. R.; Faulwasser, T.; Jones, C.; Foss, B. (PDF) Incipient Actuator Fault Handling in Nonlinear Model Predictive Control. ResearchGate. https://doi.org/10.1016/j.ifacol.2017.08.1743

[3] Cascade of Two Continuous Stirred-Tank Reactors with Recycle | Wolfram Demonstrations Project. https://demonstrations.wolfram.com/CascadeOfTwoContinuousStirredTankReactorsWithRecycle/ (accessed 2026-10-09)

[4] Sandu, A.; Verwer, J. G.; Van Loon, M.; Carmichael, G. R.; Potra, F. A.; Dabdub, D.; Seinfeld, J. H. Benchmarking Stiff Ode Solvers for Atmospheric Chemistry Problems-I. Implicit vs Explicit.

Feasibility and Scope (3-4 sentences)

At roughly 48 hours per person, a two-CSTR series system with four coupled ODEs is a good size because the model is small enough to build and debug in the first week or two, which leaves most of the time for the method comparison that is the actual point of the project. The smallest complete, gradeable version is the nonlinear four-state model at one mild operating point, solved with forward and backward Euler, explicit RK4, and BDF using scipy, and scored against a tight-tolerance reference solution on error, function evaluations, and runtime, plus a stability test showing where the explicit methods fail as the step size grows. If time allows, add a second operating point where the reactor runs hot and the reaction speeds up sharply, which makes the system stiff, hand-coded Adams-Bashforth and Adams-Moulton methods with an RK startup step, the eigenvalues of the linearized system and the stiffness ratio to explain the results, and a recycle stream that makes the coupling between the reactors two-way. Each of these extensions builds on the same model, so none of them requires restarting the project.

Risks and backup plans (1/2 page)

We are modeling two exothermic CSTRs in series with different reactions, tuned so that the system is stiff. The main risk is that the multiple steady states are harder to handle than we expect in the time we have. Following the S-shaped branch around turning points needs a continuation method, and a simple parameter sweep fails at those folds. To manage this, we will first check our steady-state code against the published cascade with identical reactions (Kubíček et al., 1980; Youssef, 1987). If full continuation is not working by the end of week 4, we will fall back to a parameter sweep from several initial guesses. We will check stability from the Jacobian eigenvalues and confirm the branches with time-domain simulations from different starting points.

A second risk is that our modified system may not show multiple steady states, or may not be stiff in the way we intend. We will start from literature parameters that do give multiplicity and move gradually toward the mixed-reaction case. We will also measure stiffness directly from the ratio of the largest to smallest Jacobian eigenvalues, and adjust the rate constants and reactor volumes until explicit methods are limited by stability rather than accuracy. If the exothermic analysis still proves too difficult, we will solve an isothermal version with nonlinear kinetics (for example A→B→C, or second-order reactions), since linear kinetics would have an analytical solution and no real numerical challenge. The project would then focus on stiffness: comparing explicit, BDF, and Radau/LSODA integrators on cost versus accuracy as we vary the reaction rates and reactor sizes. Because no published result exists for our modified system, we will validate it through the identical-reaction limit, mass and energy balance checks, and agreement with a tight-tolerance reference solution.